

A Torque-Sensor-Based Nonlinear Observer for Shockless Torque Control of Two-Inertia Systems with Backlash

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Abstract—Two-inertia resonant systems are used to represent several motion control systems with resonance modes such as drivetrains and robot joints, offering a more flexible approach for their control compared to single-inertia representations. Controlling the transmitted torsion torque (also known as joint torque) to the load allows to effectively control the driven load of the system regardless of the resonant mode, which can be done by employing torque sensors. However, many real systems contain some degree of joint backlash, which causes a large, impulse-like shock on torque reversals due to the mechanical play of the joint elements, which can lead to undesired behavior and even destruction of the mechanical elements of the system being controlled. While torque sensors are able to sense whether the drive and load-sides are in contact or not, they don't give information on the relative angle between both sides, which is necessary for achieving shockless torque control. This paper introduces a novel nonlinear observer structure based on torque sensing that is able to estimate all the load-side states even when backlash occurs. This new observer is then implemented in a torsion torque control scheme that employs model predictive control during backlash gap traversings to achieve shockless control. Simulation results confirm the effectiveness of this method.

Index Terms—Torque sensor, Nonlinear observer, Model predictive control

I. INTRODUCTION

Almost every motion control application, ranging from vehicle powertrains to robot joints and industrial equipment, exhibits mechanical resonance modes that cannot be usually ignored. In order to account for the resonance phenomenon, the two-inertia system representation has been used for their modeling and control.

One of the most important tasks within the aforementioned applications is to control the transmitted torque to the load, which in the two-inertia system representation is known as torsion torque control (also joint torque control, or shaft torque control). Methods based on load-side encoders [1] and torsion torque sensors [2] have been proposed, for which those using torsion torque sensors have a significant advantage on the fact that the controlled torsion torque is directly sensed, unlike load-side encoder-based methods where it is indirectly estimated.

However, in several cases there is some joint backlash caused by play in the mechanical parts, which causes the torque transmission characteristic of the controlled system to

become discontinuous, resulting in shocks at torque reversals which cause dangerously large peaks in the torque response. This issue has been extensively researched in the automotive field [3]. Model predictive control (MPC) has been used to achieve shockless backlash traversing [4] [5], but has the disadvantage of requiring knowledge of all the system states. Recently it has been demonstrated that using load-side encoders makes it straightforward to compensate for such effects [6].

On the other hand, almost no research has been done on the compensation of backlash when only torque sensors are available. In [7] a backlash crossing detection algorithm using torque sensors is proposed. In [8], the authors proposed a torque derivative feedback-based approach, which works well in theory, but in practice requires large current references to properly mitigate the backlash-induced shocks. This paper overcomes the limitations of this previous work by employing a backlash model and combining it with torque sensing to form a nonlinear observer capable of estimating all the load-side states even within the backlash gap. In addition to this, a torsion torque control system which incorporates MPC for shockless backlash traversing is also proposed. The effectiveness of the proposed method is demonstrated through numerical simulations.

II. CONVENTIONAL TORSION TORQUE CONTROL AND THE EFFECTS OF JOINT BACKLASH

A. Control Objective

For this paper, we consider the experimental test bench of Fig. 1 consisting of a geared motor with integrated torque sensor, a coupling with backlash which emulates the mechanical play on a real system, and a load flywheel which is to be driven.

This model can be represented as the two-inertia system of Fig. 2, where the joint backlash is modeled as a dead-zone element. The dynamics of this system are given by the following set of equations:

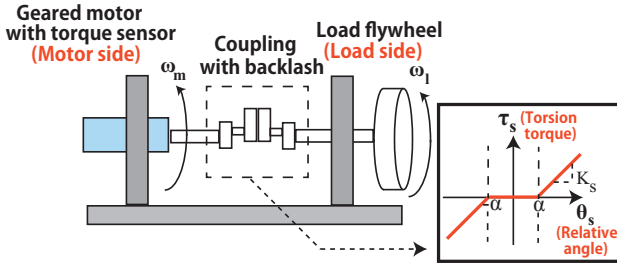


Fig. 1: Model of shaft test bench with backlash

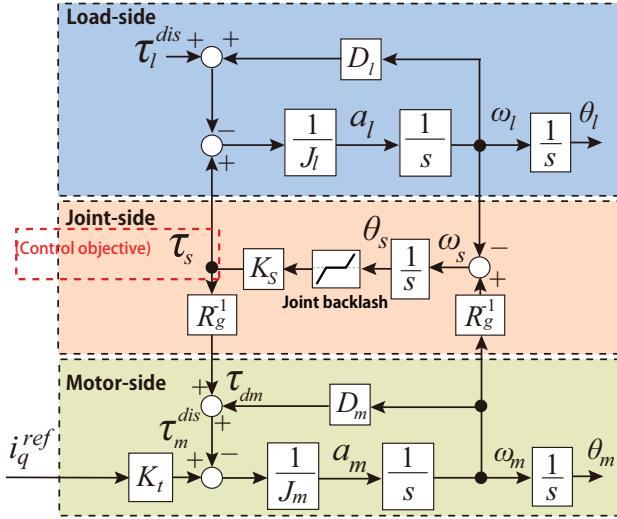


Fig. 2: Equivalent two-inertia plant and control objective

$$J_m \dot{\omega}_m = K_t i_q^{ref} - D_m \omega_m - \tau_s R_g^{-1} \quad (1)$$

$$\dot{\theta}_s = \omega_m R_g^{-1} - \omega_l \quad (2)$$

$$J_l \dot{\omega}_l = \tau_s - D_l \omega_l - \tau_l^{dis} \quad (3)$$

$$\tau_s = \begin{cases} K_s(\theta_s - \alpha), & \text{if } \theta_s \geq \alpha \\ 0, & \text{if } |\theta_s| < \alpha \\ K_s(\theta_s + \alpha), & \text{if } \theta_s \leq -\alpha \end{cases} \quad (4)$$

from which we desire to control the torsion torque τ_s . It is clear from (4) that when $|\theta_s| < \alpha$, $\tau_s = 0$ and no torque is transmitted to the load (we will refer to this plant mode as *backlash mode*). J and D are the inertia and viscous friction terms, and subscripts m and l indicate motor and load sides. K_t , K_s , R_g and α are the torque constant, spring constant, gear ratio and backlash gap half-width parameters respectively. θ , ω and a represent angular position, velocity and acceleration terms respectively, while τ^{dis} , τ_s and θ_s are the disturbance and torsion torque terms, and the relative (torsion) angle. For our setting, we consider only θ_m , ω_m and τ_s measurements to be available.

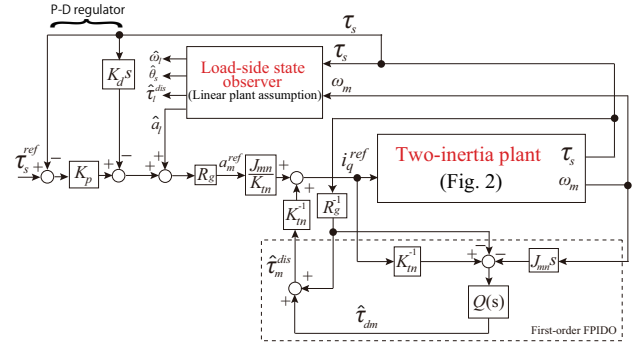


Fig. 3: Torsion torque control system under linear plant assumption

B. Torsion Torque Control System based on Linear Plant Assumption

If the backlash half-width α is small enough, the dead-zone nonlinearity can be ignored by considering $\tau_s \approx K_s \theta_s$, yielding a completely linear control system. Fig. 3 shows a torsion torque control system based on that of [9], which consists of an inner motor-side acceleration control realized by a FPIDO (Force-position integrated disturbance observer) [10], a load-side state observer and an external P-D torsion torque regulator together with load-side acceleration feedback. The load-side state observer is given as:

$$\dot{\hat{x}} = \mathbf{A}_{lin} \hat{x} + \mathbf{B}_{lin} u + \mathbf{K}(y - \hat{y}) \quad (5)$$

$$\mathbf{A}_{lin} = \begin{bmatrix} 0 & -1 & 0 \\ K_{sn} J_{ln}^{-1} & -D_{ln} J_{ln}^{-1} & -J_{ln}^{-1} \\ 0 & 0 & 0 \end{bmatrix} \quad \mathbf{B}_{lin} = \begin{bmatrix} R_g^{-1} \\ 0 \\ 0 \end{bmatrix}$$

$$\hat{x} = [\hat{\theta}_s \quad \hat{\omega}_l \quad \hat{\tau}_l^{dis}]^T \quad \mathbf{K} = [k_1 \quad k_2 \quad k_3]^T$$

$$u = \omega_m \quad y = \tau_s \quad \hat{y} = K_{sn} \hat{\theta}_s.$$

where the n subscript indicates nominal values and $\mathbf{K}$ is the observer gain vector. The load-side acceleration is estimated as:

$$\hat{a}_l = \frac{1}{J_{ln}} (\tau_s - D_{ln} \hat{\omega}_l - \hat{\tau}_l^{dis}) \quad (6)$$

The FPIDO compensates for disturbances on the motor-side, achieving motor-side acceleration control:

$$\dot{\omega}_m \approx a_m^{ref} \quad (7)$$

Then it can be demonstrated that the transfer function from the torsion torque command τ_s^{ref} to τ_s is given as:

$$\frac{\tau_s}{\tau_s^{ref}}(s) = \frac{K_{sn} K_p}{s^2 + K_d K_{sn} s + K_p K_{sn}} \quad (8)$$

C. Effects of Joint Backlash

Fig. 4 shows the simulation results of torsion torque control using the system of Fig. 3 as the value of the backlash half-width α increases. Fig. 4a shows that for larger backlash width,

unacceptably large peaks appear in the torsion torque response. On the other hand, Figs. 4b to 4d shows that the observed load-side states $[\hat{\theta}_s, \hat{\omega}_l, \hat{\tau}_l^{dis}]$ using the observer of (5) deviates from the real states when $\tau_s = 0$ and the plant is in backlash mode. $\hat{\theta}_s$ in particular never matches the real value θ_s , which is essential for achieving smooth compensation of joint backlash.

III. TORSION TORQUE-BASED NONLINEAR OBSERVER FOR LOAD-SIDE STATES ESTIMATION DURING BACKLASH

As τ_s becomes zero when the system is in backlash mode, the observability of the load states is lost, yielding incorrect observations. This fact which can be analytically demonstrated. Consider the following switching functions $H(\theta_s, \alpha)$ and $f(\theta_s, \alpha)$:

$$H(\theta_s, \alpha) = \begin{cases} 1, & \text{if } |\theta_s| \geq \alpha \\ 0, & \text{if } |\theta_s| < \alpha \end{cases} \quad (9)$$

$$f(\theta_s, \alpha) = \begin{cases} 1, & \text{if } \theta_s \geq \alpha \\ 0, & \text{if } |\theta_s| < \alpha \\ -1, & \text{if } \theta_s \leq -\alpha \end{cases} \quad (10)$$

then the joint and load-side dynamics of (2)-(4) can be rewritten as:

$$\dot{\theta}_s = \omega_m R_g^{-1} - \omega_l \quad (11)$$

$$J_l \dot{\omega}_l = \tau_s - D_l \omega_l - \tau_l^{dis} \quad (12)$$

$$\tau_s = K_s H(\theta_s, \alpha) \theta_s - K_s f(\theta_s, \alpha) \alpha \quad (13)$$

We extend the system dynamics by considering also the load-side disturbance as a system state:

$$\dot{\tau}_l^{dis} = 0, \quad (14)$$

which results in the following state-space model:

$$\dot{x} = \mathbf{A}_{nl}(\theta_s)x + \mathbf{B}_{nl}u - \mathbf{P}f(\theta_s, \alpha) \quad (15)$$

$$y = \tau_s = \mathbf{C}_{nl}(\theta_s)x + \mathbf{D}_{nl} \quad (16)$$

$$\mathbf{A}_{nl}(\theta_s) = \begin{bmatrix} 0 & -1 & 0 \\ K_s J_l^{-1} H(\theta_s, \alpha) & -D_l J_l^{-1} & -J_l^{-1} \\ 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{B}_{nl} = [R_g^{-1} \ 0 \ 0]^T \quad \mathbf{P} = [0 \ K_s J_l^{-1} \alpha \ 0]^T$$

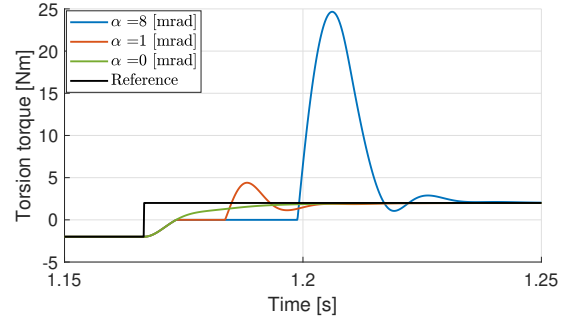
$$\mathbf{C}_{nl} = [K_s H(\theta_s, \alpha) \ 0 \ 0] \quad \mathbf{D}_{nl} = [-K_s \alpha f(\theta_s, \alpha)]$$

$$x = [\theta_s \ \omega_l \ \tau_l^{dis}]^T \quad u = \omega_m$$

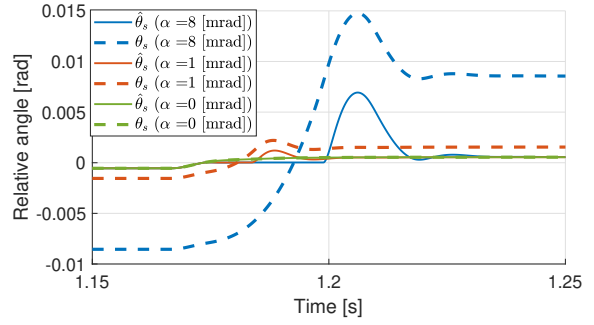
We can derive the observability matrix $\mathcal{O} = [\mathbf{C}_{nl} \ \mathbf{C}_{nl}\mathbf{A}_{nl} \ \mathbf{C}_{nl}\mathbf{A}_{nl}^2]^T$ as:

$$\mathcal{O}(\theta_s) = \begin{bmatrix} K_s H(\theta_s, \alpha) & 0 & 0 \\ 0 & -K_s H(\theta_s, \alpha) & 0 \\ -\frac{K_s^2}{J_l} H(\theta_s, \alpha)^2 & \frac{K_s}{J_l} D_l H(\theta_s, \alpha) & \frac{K_s}{J_l} H(\theta_s, \alpha) \end{bmatrix} \quad (17)$$

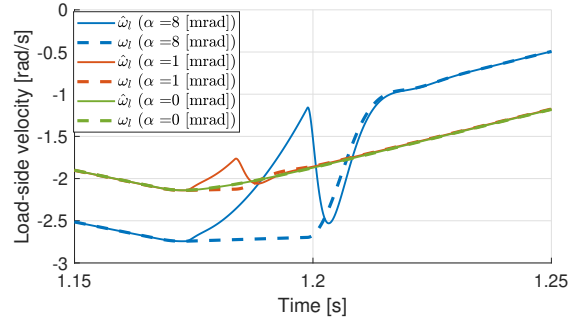
and during backlash mode we have that:



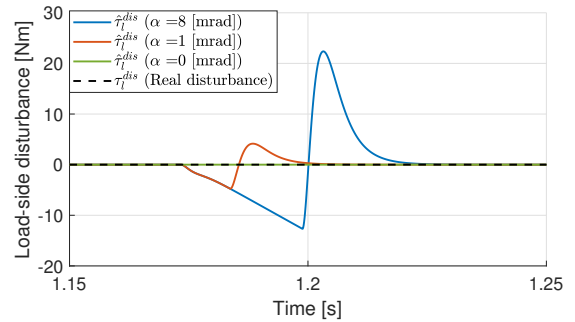
(a) Torsion torque τ_s



(b) Relative angle θ_s observation results



(c) Load side velocity ω_l observation results



(d) Load side disturbance τ_l^{dis} observation results

Fig. 4: Results for torsion torque control simulations on different backlash widths during torque reversals (torsion torque is driven from -2 to 2 Nm)

$$\mathcal{O}(-\alpha < \theta_s < \alpha) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (18)$$

which indicates the impossibility of observing the load-side states using torque sensing when the system is in backlash mode.

A. Forming the Nonlinear Observer

Given that it is impossible to observe the load-side states during backlash mode, one alternate approach is to instead predict them by using a virtual model of the plant during backlash, in a similar fashion to [5]. The process of switching between observation and estimation can be neatly integrated into a single nonlinear observer of the following form:

$$\dot{\hat{x}} = \bar{\mathbf{A}}_{\text{nl}}(\hat{\theta}_s)\hat{x} + \bar{\mathbf{B}}_{\text{nl}}u + \mathbf{K}(y - \hat{y}) - \bar{\mathbf{P}}f(\hat{\theta}_s, \alpha_n) \quad (19)$$

$$y = \tau_s = \mathbf{C}_{\text{nl}}(\theta_s)x + \mathbf{D}_{\text{nl}} \quad (20)$$

$$\hat{y} = \hat{\tau}_s = \bar{\mathbf{C}}_{\text{nl}}(\hat{\theta}_s)\hat{x} + \bar{\mathbf{D}}_{\text{nl}} \quad (21)$$

$$\bar{\mathbf{A}}_{\text{nl}}(\theta_s) = \begin{bmatrix} 0 & -1 & 0 \\ K_{sn}J_{ln}^{-1}H(\hat{\theta}_s, \alpha_n) & -D_{ln}J_{ln}^{-1} & -J_{ln}^{-1} \\ 0 & 0 & 0 \end{bmatrix}$$

$$\bar{\mathbf{B}}_{\text{nl}} = [R_g^{-1} \ 0 \ 0]^T \quad \bar{\mathbf{P}} = [0 \ K_{sn}J_{ln}^{-1}\alpha_n \ 0]^T$$

$$\bar{\mathbf{C}}_{\text{nl}} = [K_{sn}H(\hat{\theta}_s, \alpha_n) \ 0 \ 0] \quad \bar{\mathbf{D}}_{\text{nl}} = [-K_{sn}\alpha_n f(\hat{\theta}_s, \alpha_n)]$$

$$\hat{x} = [\hat{\theta}_s \ \hat{\omega}_l \ \hat{\tau}_l^{dis}]^T \quad u = \omega_m$$

The feedback error term e is given as:

$$e = y - \hat{y} = \tau_s - \hat{\tau}_s$$

$$= K_s H(\theta_s, \alpha) - K_s \alpha f(\theta_s, \alpha)$$

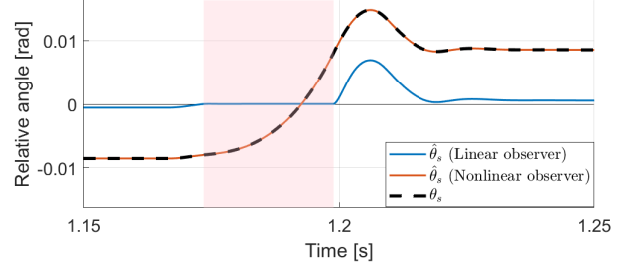
$$- K_{sn} H(\hat{\theta}_s, \alpha_n) + K_{sn} \alpha_n f(\hat{\theta}_s, \alpha_n). \quad (22)$$

Note that this feedback error uses the torque sensor information τ_s , which itself contains the real dynamics of the joint backlash (as shown in (20)), and $\hat{\tau}_s$, which is formed by a virtual model of the joint backlash based on the estimate relative angle $\hat{\theta}_s$ (as given by (21)). It is clear from this equation that a good estimation of the nominal spring constant K_{sn} and nominal backlash half-width α_n is needed. If this condition is met and $K_{sn} \approx K_s$, $\alpha_n \approx \alpha$, then e is rewritten as:

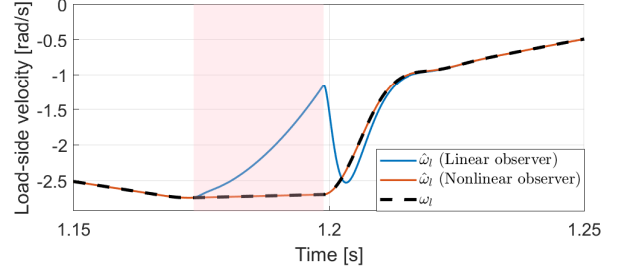
$$e = K_{sn}(H(\theta_s, \alpha) - H(\hat{\theta}_s, \alpha_n))$$

$$+ K_{sn}\alpha_n(f(\hat{\theta}_s, \alpha_n) - f(\theta_s, \alpha)). \quad (23)$$

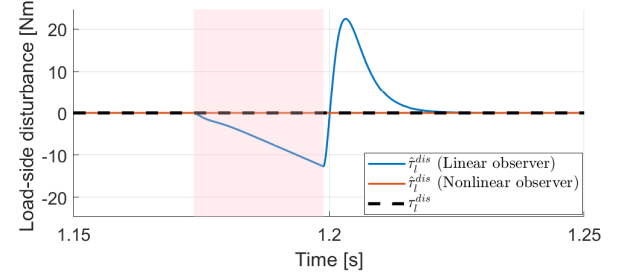
This rewritten form tells us that it is also necessary for $\hat{\theta}_s$ to converge rapidly enough to θ_s in order to minimize the error between the nominal and actual backlash models $[H(\hat{\theta}_s, \alpha_n), f(\hat{\theta}_s, \alpha_n)]$ and $[H(\theta_s, \alpha), f(\theta_s, \alpha)]$. This is important in particular to accurately predict the load-side states during backlash mode. If e quickly converges to zero during backlash mode, then the observer of (19) takes the following form:



(a) Torsion angle θ_s observation results



(b) Load side velocity ω_l observation results



(c) Load side disturbance τ_l^{dis} observation results

Fig. 5: Simulation results using proposed nonlinear observer. Pink highlighted zone indicates that two-inertia system is in backlash mode.

$$\dot{\hat{x}} = \bar{\mathbf{A}}_{\text{nl}}(\hat{\theta}_s)\hat{x} + \bar{\mathbf{B}}_{\text{nl}}u - \bar{\mathbf{P}}f(\hat{\theta}_s, \alpha_n), \quad (24)$$

and the observer works instead as a predictor (simulator) of the load-side dynamics during backlash mode. Fig. 5 shows the simulation results for both the linear observer of (5) and the proposed nonlinear observer of (19), demonstrating that the proposed observer is capable of estimating the load-side states even during backlash mode.

IV. SHOCKLESS TORSION TORQUE CONTROL SCHEME USING MODEL PREDICTIVE CONTROL

As all the load-side states are able to be observed/estimated even during backlash, it is possible to use MPC to achieve quick and shockless backlash gap traversals. MPC for drive-trains with backlash has been applied in previous research for backlash-induced shock mitigation [4] [5].

A. MPC Formulation for Backlash Gap Traversals

As the system described by (1)-(4) is nonlinear, convexity of the quadratic program to be solved in the MPC scheme cannot

be guaranteed. Thus, we use only the plant model for backlash mode when $|\theta_s| < \alpha$. Referring to the deadzone model of (4), for this condition:

$$\tau_s = 0 \quad (25)$$

thus, the two-inertia plant dynamics during backlash are given as:

$$\dot{\omega}_m = \frac{K_t}{J_m} i_q^{ref} - \frac{D_m}{J_m} \quad (26)$$

$$\dot{\theta}_s = \omega_m R_g^{-1} - \omega_l \quad (27)$$

$$\dot{\omega}_l = -\frac{D_l}{J_l} \omega_l - \frac{1}{J_l} \tau_l^{dis} \quad (28)$$

We once again form an extended plant considering the disturbance term τ_l^{dis} as an additional state:

$$\dot{x} = \mathbf{A}_{\text{mpc}} x + \bar{\mathbf{B}}_{\text{mpc}} u \quad (29)$$

$$y = \theta_s = [0 \ 1 \ 0 \ 0] x \quad (30)$$

$$\mathbf{A}_{\text{mpc}} = \begin{bmatrix} -\frac{D_m}{J_m} & 0 & 0 & 0 \\ R_g^{-1} & 0 & -1 & 0 \\ 0 & 0 & -\frac{D_l}{J_l} & -\frac{1}{J_l} \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \bar{\mathbf{B}}_{\text{mpc}} = \begin{bmatrix} \frac{K_t}{J_m} \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (31)$$

$$x = [\omega_m \ \theta_s \ \omega_l \ \tau_l^{dis}]^T \quad u = i_q^{ref} \quad (32)$$

This plant model is then used to formulate the MPC controller. We use two different formulations, one for positive traversals ($\tau_s^{ref} > 0$):

$$\min_{u_i} \sum_{k=i}^{i+H_p} w_y \|y_k - y_+^{ref}\|_2^2 + \sum_{k=i}^{i+H_p-1} w_{\Delta u} \|\Delta u_k\|_2^2 \quad (33)$$

$$\begin{aligned} \text{s.t. } & x_{k+1} = \mathbf{A}_d x_k + \mathbf{B}_d u_k \\ & \Delta u_k = 0 \quad \forall k \in \{i + H_c, \dots, i + H_p - 1\} \\ & u^{min} \leq u_k \leq u^{max} \quad \forall k \in \{i, \dots, i + H_p - 1\} \\ & \theta_{s_j} \leq \alpha - \epsilon \quad \forall j \in \{i, \dots, i + H_p\}, \end{aligned}$$

and another for negative traversals ($\tau_s^{ref} < 0$):

$$\min_{u_i} \sum_{k=i}^{i+H_p} w_y \|y_k - y_-^{ref}\|_2^2 + \sum_{k=i}^{i+H_p-1} w_{\Delta u} \|\Delta u_k\|_2^2 \quad (34)$$

$$\begin{aligned} \text{s.t. } & x_{k+1} = \mathbf{A}_d x_k + \mathbf{B}_d u_k \\ & \Delta u_k = 0 \quad \forall k \in \{i + H_c, \dots, i + H_p - 1\} \\ & u^{min} \leq u_k \leq u^{max} \quad \forall k \in \{i, \dots, i + H_p - 1\} \\ & \epsilon - \alpha \leq \theta_{s_j} \quad \forall j \in \{i, \dots, i + H_p\} \end{aligned}$$

where:

$$y_+^{ref} = \alpha - \epsilon \quad (35)$$

$$y_-^{ref} = \epsilon - \alpha, \quad (36)$$

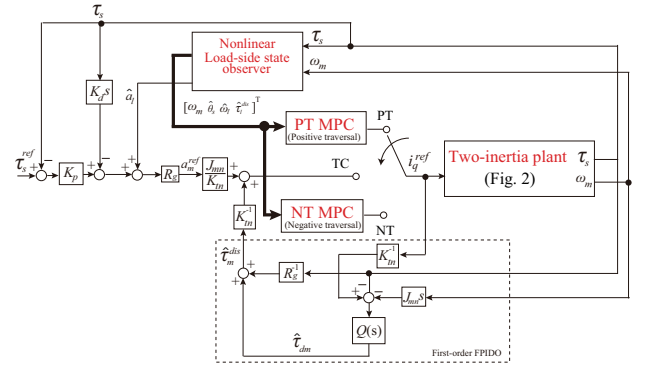


Fig. 6: Proposed torsion torque control system.

TABLE I: Simulation parameters

Parameter	Symbol	Value
FPIDO bandwidth		500 rad/s
Nonlinear observer bandwidth		500 rad/s
Simulation sample time	T_s	100 μ s
MPC sampling time	T_{mpc}	4 ms
MPC prediction horizon	H_p	8
MPC control horizon	H_c	4
MPC tracking cost	w_y	1
MPC input rate cost	$w_{\Delta u}$	0.001

where ϵ is a very small constant. That is, we design the MPC controllers and its references so that the backlash gap is almost, but not completely traversed, approach introduced in [4]. H_p , H_c , w_y and $w_{\Delta u}$ are prediction horizon, control horizon, reference tracking cost and input rate-of-change cost respectively. The inequality constraints represent constraints on i_q^{ref} and on the relative angle θ_s . $\mathbf{A}_d$ and $\mathbf{B}_d$ are the discrete-time representations of $\mathbf{A}_{\text{mpc}}$ and $\mathbf{B}_{\text{mpc}}$.

B. Proposed Control Structure

The proposed control structure is summarized in Fig. 6. The control mode is switched between the PT (positive traversal, (33)), TC (torque control) and NT (negative traversal, (34)) modes, based on the switching law of (37).

Control mode: (37)

PT if $\tau_s^{ref} > 0 \quad \wedge \quad \hat{\theta}_s \leq y_+^{ref} \quad \wedge \quad |\tau_s| \leq \gamma$

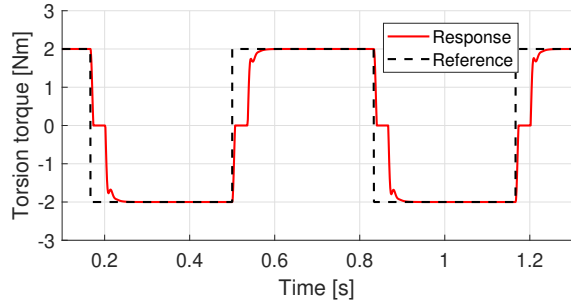
NT if $\tau_s^{ref} < 0 \quad \wedge \quad \hat{\theta}_s \geq y_-^{ref} \quad \wedge \quad |\tau_s| \leq \gamma$

TC if else

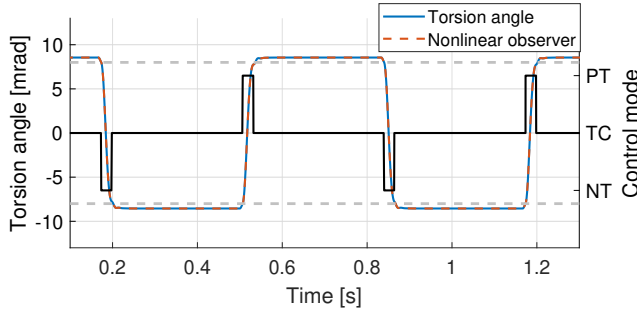
Note that the PT and NT modes are only activated on either positive or negative reversals, with the additional escape condition $|\tau_s| \leq \gamma$ applied to avoid false positives when $\hat{\theta}_s \neq \theta_s$, where γ is a sufficiently small value used to detect whether the system is really in backlash mode or not by using τ_s information. TC mode uses the same linear controller as Fig. 3.

V. SIMULATION SETTING AND RESULTS

Table I shows the simulation parameters. ϵ and γ are set to $0.03 \times \alpha_n$ and 0.01 Nm, respectively, where α_n is the nom-



(a) Torsion torque τ_s



(b) Relative angle θ_s and controller switching (dashed lines indicate the backlash width $[-\alpha, \alpha]$)

Fig. 7: Simulation results of proposed torsion torque controller

inal backlash half-width. Simulations were run on MATLAB Simulink, and the MPC controllers were implemented using the Model Predictive Control Toolbox.

Fig. 7 shows the simulation results of the proposed torsion torque controller, where it is confirmed that shockless torsion torque control is achieved. It is confirmed from Fig. 7b that the proposed nonlinear observer is capable of estimating θ_s , allowing to correctly switch between the different system modes.

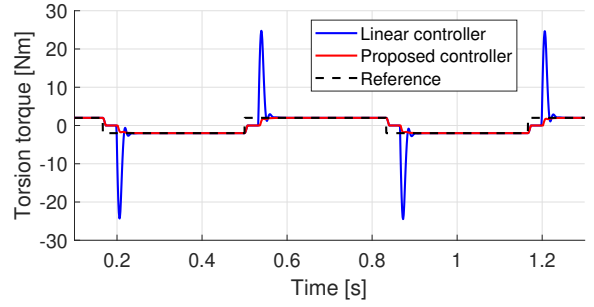
A comparison with the linear controller of Fig. 3 is shown in Fig. 8, where the overshoot of nearly 25 Nm of the conventional method was completely suppressed while also improving convergence time.

VI. CONCLUSION

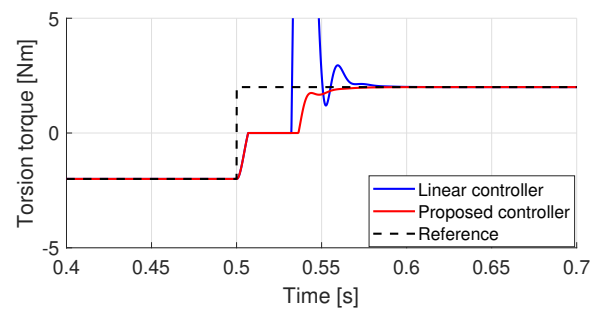
This paper presented a torsion torque controller for two-inertia systems with backlash by a novel nonlinear observer based on torque sensing and model predictive control. Simulation results demonstrated that the proposed observer yielded accurate estimations of the two-inertia system load-side states even during backlash mode thus achieving shockless torque control by applying MPC on backlash traversals. Experimental verification of the proposed methodology on a real test bench and its robustness analysis will be done in future work.

ACKNOWLEDGMENT

This work was partially supported by Tsugawa Foundation and JSPS KAKENHI Grant Number JP24924938.



(a) Torsion torque τ_s



(b) Zoomed-in view

Fig. 8: Comparison of conventional linear controller with proposed method.

REFERENCES

- [1] B. Ugurlu, E. Sariyildiz, A. T. Kansizoglu, E. C. Ozcinar and S. Coruk: "Benchmarking Torque Control Strategies for a Torsion-Based Series Elastic Actuator," in *IEEE Robotics & Automation Magazine*, vol. 29, no. 2, pp. 85-96, June 2022, doi: 10.1109/MRA.2021.3124154.
- [2] Y. Yokokura and K. Ohishi: "Fine Load-Side Acceleration Control Based on Torsion Torque Sensing of Two-Inertia System," in *IEEE Transactions on Industrial Electronics*, vol. 67, no. 1, pp. 768-777, Jan. 2020, doi: 10.1109/TIE.2018.2881944.
- [3] M. Nordin and P. Gutman: "Controlling mechanical systems with backlash—A survey," in *Automatica*, vol. 38, no. 10, pp. 1633-1649, 2002. 49.
- [4] A. Lagerberg and B. Egardt, "Model predictive control of automotive powertrains with backlash," in *Proc. 16th IFAC World Congress*, Prague, Czech Republic, 2005, pp. 1-6.
- [5] A. Formentini, A. Oliveri, M. Marchesoni and M. Storace, "A Switched Predictive Controller for an Electrical Powertrain System With Backlash," in *IEEE Transactions on Power Electronics*, vol. 32, no. 5, pp. 4036-4047, May 2017, doi: 10.1109/TPEL.2016.2587756
- [6] S. Yamada and H. Fujimoto: "Precise Joint Torque Control Method for Two-inertia System with Backlash Using Load-side Encoder," in *IEEE Journal of Industry Applications*, vol. 8, no. 1, pp. 75-83, 2019.
- [7] D. Yanakiev *et al*, "Lash crossing detection using a shaft torque sensor," US Patent 8733183B1, 2014.
- [8] J. Padron, Y. Kawai, Y. Yokokura, K. Ohishi and T. Miyazaki, "A Method for Gear Impact Suppression In Torsion Torque Control of Two-Inertia Systems with Backlash," 2021 IEEE 30th International Symposium on Industrial Electronics (ISIE), Kyoto, Japan, 2021, pp. 1-6
- [9] J. Padron, T. Miyazaki, Y. Yokokura and K. Ohishi, "Integratorless 2DOF Torsion Torque Control for Geared Motors with Transmission Error," IECON 2023- 49th Annual Conference of the IEEE Industrial Electronics Society, Singapore, Singapore, 2023, pp. 1-6, doi: 10.1109/IECON51785.2023.10312702.
- [10] Y. Kawai, Y. Yokokura, K. Ohishi and P. Boonwong, "High-robust acceleration control using force and position sensors integrated disturbance observer," IECON 2016 - 42nd Annual Conference of the IEEE Industrial Electronics Society, Florence, Italy, 2016, pp. 5802-5807